A Half-Quadratic Threshold for Volume Monotonicity of Star-Shaped Minkowski Sums

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Abstract

For a compact star-shaped set $S \subset \mathbb{R}^d$, Fradelizi, Lángi, and Zvavitch proved

$$\operatorname{vol}\left(\frac{S[k+1]}{k+1}\right) \geq \operatorname{vol}\left(\frac{S[k]}{k}\right) \qquad \text{when } k \geq (d-1)(d-2).$$

We show that their argument in fact proves the same conclusion, with the same equality characterization, whenever

$$(k-d+2)(k+1)^{d-1} > k^d, \qquad k \geq d-1.$$

In particular it is enough that

$$k \geq \binom{d}{2} - 1.$$

This replaces a threshold asymptotic to d^2 by one asymptotic to $d^2/2$. The exact scalar condition is slightly stronger still: its first valid values for $d = 4, 5, 6, 7$ are respectively $k = 5, 8, 13, 19$, whereas the published thresholds are $6, 12, 20, 30$.

1 Source problem and scope

For a set $A \subset \mathbb{R}^d$, write

$$A[k] = \underbrace{A + \cdots + A}_{k \text{ summands}}.$$

The Bobkov–Madiman–Wang conjecture asks whether $\operatorname{vol}(A[k]/k)$ is nondecreasing for every compact A. The conjecture is false in sufficiently high dimension, but important structured cases remain open. The source paper proves the star-shaped case for $d = 2, 3$ and, in arbitrary dimension, for $k \geq (d-1)(d-2)$; its Conjecture 1, definition, and Theorem 1 appear together on page 2 in Figure 1.

Our result does not settle the remaining low-k star-shaped cases, nor the general conjecture. It is a uniform strengthening of the source's positive range in every dimension $d \geq 4$.

2 Proof intuition

The source proof decomposes a polytope into simplicial cones and reduces the problem to adjoining a union B of d radial segments to a set M trapped between $B[k]$ and $k \operatorname{conv}(B)$. Its cubical-layer transport estimate gives the stronger affine inequality

$$\operatorname{vol}\left(\frac{M+B}{k+1}\right) \geq \alpha_{d,k} \operatorname{vol}\left(\frac{M}{k}\right) + (1-\alpha_{d,k}) \operatorname{vol}(\operatorname{conv} B), \qquad \alpha_{d,k} = \frac{k^d}{(k-d+2)(k+1)^{d-1}}.$$

A further step in the investigation of the sequence $\left\{\frac{1}{k}A[k]\right\}$ is to examine the monotonicity of this convergence. Whereas this sequence is clearly not monotonous in terms of containment, the main object of this paper is the following conjecture of Bobkov, Madiman, Wang [BMW], relating the volumes of the elements of the sequence, and in which $\mathrm{vol}(K)$ denotes the Lebesgue measure (volume) of the measurable set $K \subset \mathbb{R}^d$.

Conjecture 1 (Bobkov-Madiman-Wang)**.** Let A be a compact set in $\mathbb{R}^d$ for some $d \in \mathbb{N}$. Then the sequence

$$\left\{\mathrm{vol}\left(\frac{1}{k}A[k]\right)\right\}_{k\geq 1}$$

is non-decreasing in k.

Equivalently, this conjecture asks whether for any integer $k \geq 1$ and compact set $A \subset \mathbb{R}^d$, the following inequality holds

$$\mathrm{vol}\left(\frac{1}{k}A[k]\right) \leq \mathrm{vol}\left(\frac{1}{k+1}A[k+1]\right). \tag{1}$$

This inequality trivially holds for any compact set A if $k=1$ since $A \subset \frac{1}{2}A[2]$. In the same way, it is easy to find monotone subsequences of the sequence $\{\mathrm{vol}(\frac{1}{k}A[k])\}_{k\geq 1}$ by the same argument; one such example is $\{\mathrm{vol}(\frac{1}{2^m}A[2^m])\}_{m\geq 0}$. On the other hand, even the first nontrivial case; that is, the inequality $\mathrm{vol}\left(\frac{1}{2}A[2]\right) \leq \mathrm{vol}\left(\frac{1}{3}A[3]\right)$ seems to require new methods to approach. Conjecture 1 was partially resolved in [FMMZ1, FMMZ2], where, following the approach of [GMR], the authors proved it for any 1-dimensional compact set A, but constructed counterexamples in $\mathbb{R}^d$ for any $d \geq 12$. More precisely, they showed that for every $k \geq 2$, there is $d_k \in \mathbb{N}$ such that for every $d \geq d_k$ there is a compact set $A \subset \mathbb{R}^d$ such that $\mathrm{vol}\left(\frac{1}{k}A[k]\right) > \mathrm{vol}\left(\frac{1}{k+1}A[k+1]\right)$. In particular, one has $d_2 = 12$, whence Conjecture 1 fails for $\mathbb{R}^d$ if $d \geq 12$.

Our goal is to find additional conditions on A and k under which the statement in Conjecture 1, or more precisely when the inequality (1) is satisfied.

In the paper, for any set $A \subset \mathbb{R}^d$ we denote by $\dim A$ the dimension of the smallest affine subspace containing A, and for any $p, q \in \mathbb{R}^d$, we denote the closed segment with endpoints p, q by $[p, q]$. To state our main result, let us recall the following well-known concept.

Definition 1. A nonempty set $S \subset \mathbb{R}^d$ is called *star-shaped* with respect to a point p if for any $q \in S$, we have $[p, q] \subseteq S$.

Our main result is the following.

Theorem 1. *Let $d \geq 2$ and $k \geq \max\{2, (d-1)(d-2)\}$ be integers. Then for any compact, star-shaped set $S \subset \mathbb{R}^d$ we have*

$$\mathrm{vol}\left(\frac{1}{k+1}S[k+1]\right) \geq \mathrm{vol}\left(\frac{1}{k}S[k]\right),$$

with equality if only if $\dim(S) < d$ or $\frac{1}{k}S[k] = \mathrm{conv}(S)$.

Figure 1: Source page 2: the BMW volume-monotonicity conjecture and the published star-shaped threshold, arXiv:1910.06146v2.

Only the last line of that proof replaces the exact requirement $\alpha_{d,k} < 1$ by the convenient sufficient bound $k \geq (d-1)(d-2)$. Retaining the exact coefficient immediately enlarges the range. A second-order Taylor estimate shows that $\alpha_{d,k} < 1$ already when $k+1 \geq \binom{d}{2}$. No geometric part of the argument changes.

3 The sharpened theorem

Theorem 1 (Exact coefficient range)**.** *Let $d \geq 3$ and $k \geq \max\{2, d-1\}$ be integers satisfying*

$$(k-d+2)(k+1)^{d-1} > k^d. \tag{1}$$

Then every compact star-shaped set $S \subset \mathbb{R}^d$ satisfies

$$\operatorname{vol}\left(\frac{S[k+1]}{k+1}\right) \geq \operatorname{vol}\left(\frac{S[k]}{k}\right). \tag{2}$$

Equality holds if and only if either $\dim S < d$ or $S[k]/k = \operatorname{conv} S$.

Corollary 2 (Half-quadratic closed form)**.** *The conclusion of Theorem 1 holds for every $d \geq 3$ and*

$$k \geq \binom{d}{2} - 1.$$

Thus the published condition $(d-1)(d-2)$ can be replaced by $\binom{d}{2} - 1$.

It is useful to distinguish the exact cutoff for the coefficient (1) from a possibly smaller threshold obtainable by a different geometric method. Define

$$\kappa_d = \min\{k \geq d-1 : (k-d+2)(k+1)^{d-1} > k^d\}.$$

Exact integer arithmetic gives:

d	3	4	5	6	7	8	9	10	11	12
κ_d	2	5	8	13	19	25	33	42	51	62
$\binom{d}{2} - 1$	2	5	9	14	20	27	35	44	54	65
$(d-1)(d-2)$	2	6	12	20	30	42	56	72	90	110

4 Local coefficient lemma

Let $p_1, \dots, p_d$ be a basis of $\mathbb{R}^d$, let the common endpoint be the origin, and put

$$B = \bigcup_{i=1}^{d} [0, p_i], \qquad V = \operatorname{vol}(\operatorname{conv} B).$$

Lemma 3 (Range-sharpened local lemma)**.** *Assume $k \geq \max\{2, d-1\}$, $B[k] \subset M \subset k \operatorname{conv} B$, and M is compact. Set*

$$\alpha_{d,k} = \frac{k^d}{(k-d+2)(k+1)^{d-1}}.$$

Then

$$\operatorname{vol}\left(\frac{M+B}{k+1}\right) \geq \alpha_{d,k} \operatorname{vol}\left(\frac{M}{k}\right) + (1-\alpha_{d,k})V. \tag{3}$$

If $\alpha_{d,k} < 1$, then

$$\mathrm{vol}\left(\frac{M+B}{k+1}\right) \geq \mathrm{vol}\left(\frac{M}{k}\right),$$

with equality only for $M = k \operatorname{conv} B$. Moreover, if the difference between the left and right normalized volumes is at most δ, then

$$\mathrm{vol}(k \operatorname{conv} B) - \mathrm{vol}(M) \leq \frac{k^d}{1-\alpha_{d,k}}\,\delta. \tag{4}$$

Proof. Equation (3) is exactly inequality (10) in the source paper [1], before its final scalar estimate. We record why the larger range causes no hidden boundary problem. The proof partitions $k \operatorname{conv} B$ into unit cubical layers indexed from $t = k-d+1$ to $k-1$. Every preceding counting and mass-transport identity is valid as soon as $k-d+1 \geq 0$; at the endpoint $k = d-1$, the first layer is simply $t = 0$. The only later use of $k \geq (d-1)(d-2)$ is the displayed binomial lower bound intended to make $1-\alpha_{d,k}$ nonnegative. Thus the derivation of (3) itself is valid for $k \geq d-1$.

Write $u = \mathrm{vol}(M/k)$. Since $M \subset k \operatorname{conv} B$, one has $u \leq V$, and (3) becomes

$$\mathrm{vol}\left(\frac{M+B}{k+1}\right) \geq u + (1-\alpha_{d,k})(V-u).$$

When $\alpha_{d,k} < 1$, this proves monotonicity. Equality forces $u = V$. Hence $k \operatorname{conv} B \setminus M$ has measure zero. The compactness of M and the fact that a full-dimensional compact convex body is the closure of its interior imply $M = k \operatorname{conv} B$. The same displayed inequality gives $(1-\alpha_{d,k})(V-u) \leq \delta$; multiplication by k^d proves (4). □

5 Proof of the global theorem

Proof of Theorem 1. Condition (1) is precisely $\alpha_{d,k} < 1$, so Lemma 3 applies. Translate a star center of S to the origin. The case $\dim S < d$ is immediate, so assume S is full-dimensional.

Fix $\varepsilon > 0$. Choose a finite set $A_0 \subset S$, enlarged to contain the origin, with

$$\mathrm{vol}(\operatorname{conv} S) - \mathrm{vol}(\operatorname{conv} A_0) \leq \varepsilon.$$

Triangulate the boundary of $\operatorname{conv} A_0$. Coning its facets to the origin gives simplices $T_j = \operatorname{conv} B_j$ with mutually disjoint interiors, where each B_j is the union of the d radial segments from the origin to the vertices of the corresponding boundary simplex. Star-shapedness gives $B_j \subset S$. Put

$$M_j = S[k] \cap kT_j.$$

Then $B_j[k] \subset M_j \subset kT_j$, so Lemma 3 yields

$$\mathrm{vol}\left(\frac{M_j+B_j}{k+1}\right) \geq \mathrm{vol}\left(\frac{M_j}{k}\right).$$

The sets on the left lie in $S[k+1]/(k+1)$ and have disjoint interiors. Since $S[k]/k \subset \operatorname{conv} S$, the portion of $S[k]/k$ outside $\operatorname{conv} A_0$ has volume at most ε. Summing over j gives

$$\mathrm{vol}\left(\frac{S[k]}{k}\right) - \varepsilon \leq \sum_j \mathrm{vol}\left(\frac{M_j}{k}\right) \leq \sum_j \mathrm{vol}\left(\frac{M_j+B_j}{k+1}\right) \leq \mathrm{vol}\left(\frac{S[k+1]}{k+1}\right).$$

Letting $\varepsilon \downarrow 0$ proves (2).

For the equality case, suppose the two endpoint volumes are equal. Let δ_j be the nonnegative difference between the two normalized local volumes. The preceding chain gives $\sum_j \delta_j \le \varepsilon$. By (4), with $C_{d,k} = k^d/(1-\alpha_{d,k})$,

$$\operatorname{vol}(k \operatorname{conv} A_0) - \operatorname{vol}(S[k] \cap k \operatorname{conv} A_0) \le C_{d,k}\varepsilon.$$

The part of $k \operatorname{conv} S$ outside $k \operatorname{conv} A_0$ has volume at most $k^d\varepsilon$. Consequently

$$\operatorname{vol}(k \operatorname{conv} S) - \operatorname{vol}(S[k]) \le (k^d + C_{d,k})\varepsilon.$$

Letting $\varepsilon \downarrow 0$ and using compactness as in the local lemma gives $S[k] = k \operatorname{conv} S$. Conversely, if this set equality holds, the left normalized volume is already $\operatorname{vol}(\operatorname{conv} S)$; monotonicity and the containment $S[k+1]/(k+1) \subset \operatorname{conv} S$ force equality. This proves the stated characterization. $\square$

6 The half-quadratic scalar estimate

Proof of Corollary 2. Put $x = k+1$ and $t = 1/x$. Condition (1) is equivalent to

$$(1-t)^d < 1 - (d-1)t. \tag{5}$$

For $d \ge 3$ and $0 < t < 1$, Taylor's theorem at zero gives the strict upper bound

$$(1-t)^d < 1 - dt + \binom{d}{2} t^2,$$

because the third derivative of $(1-t)^d$ is negative. If $x \ge \binom{d}{2}$, then $\binom{d}{2} t \le 1$, and hence

$$1 - dt + \binom{d}{2} t^2 \le 1 - dt + t = 1 - (d-1)t.$$

This proves (5), hence (1). Since $k \ge \binom{d}{2} - 1$ also implies $k \ge d-1$, Theorem 1 applies. $\square$

For completeness, the exact coefficient cutoff is unambiguous. The function

$$h_d(t) = 1 - (d-1)t - (1-t)^d$$

has $h_d(0) = 0$, $h_d'(0) = 1$, and $h_d''(t) = -d(d-1)(1-t)^{d-2} < 0$ on $(0,1)$. Thus it has at most one positive zero, and the positivity test can change only once as k increases and $t = 1/(k+1)$ decreases. The table above therefore records exact cutoffs for this coefficient argument, not a collection of isolated successes.

7 Verification, novelty, and limitations

The reusable checker in `code/verify_thresholds.py` uses exact integer arithmetic. It verifies the displayed values of κ_d, confirms strict positivity at the closed-form threshold through $d = 1000$, and checks the equivalence between the polynomial and normalized forms of the scalar condition through $d = 200$. These computations are sanity checks; the proof of the uniform statement is the Taylor argument above.

The main human-review point is narrow: confirm that the cubical-layer derivation of source inequality (10) uses only $k \geq d-1$ before the final coefficient estimate. The source text has been audited term by term, including the endpoint layer $t = 0$.

For novelty, the run's solution, attempt, and proof-gap indexes were searched for arXiv:1910.06146 and the star-shaped threshold. Focused arXiv searches on 9 August 2026 used the exact paper title, the scalar factor `(k-d+2)(k+1)`, and combinations of "star-shaped", "Minkowski sum", "volume monotonicity", and "threshold". They found the source paper [1], the earlier general counterexample paper [2], and broader subset-sum work, but no later paper stating this refinement. This is a bounded novelty check, not a claim of exhaustive bibliographic priority.

The packet is a substantial partial result. It does not resolve the remaining star-shaped cases $k < \kappa_d$; for example, dimensions four and five still have the unresolved ranges $k = 2, 3, 4$ and $k = 2, 3, 4, 5, 6, 7$, respectively.

References

[1] M. Fradelizi, Z. Lángi, and A. Zvavitch, *Volume of the Minkowski sums of star-shaped sets*, arXiv:1910.06146v2 (2021), `https://arxiv.org/abs/1910.06146`.

[2] M. Fradelizi, M. Madiman, A. Marsiglietti, and A. Zvavitch, *Do Minkowski averages get progressively more convex?*, arXiv:1512.03718 (2015), `https://arxiv.org/abs/1512.03718`.